



Optical Absorption and EPR Studies on Gamma-Ray Irradiated RE³⁺-Doped Fluorophosphate Glasses

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Received: 24 July 2017 / Accepted: 27 January 2018 / Published online: 19 February 2018
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Abstract

We have studied the effect of γ -ray irradiation on optical absorption, emission and decay characteristics of RE³⁺ (RE = Sm, Eu and Dy)-doped fluorophosphate glasses. Electron paramagnetic resonance (EPR) study confirms the POHC and PO₃ EC defects induced in glasses by the γ -irradiation. The presence of induced defect centers significantly affects the optical and emission properties. The optical band gap values of the studied systems increased after the γ -ray irradiation. The phonon energy and electron–phonon coupling strength of Eu³⁺-doped fluorophosphate glass were determined from the phonon sideband analysis. The emission intensity of the RE³⁺ ions increased significantly after the γ -ray irradiation. The intensity parameter, R is the ratio of the intensities of the $^5D_0 \rightarrow ^7F_2/^5D_0 \rightarrow ^7F_1$ transitions of Eu³⁺ ion and Y/B intensity parameter is the ratio of intensities of the $^4F_{9/2} \rightarrow ^6H_{13/2}/^4F_{9/2} \rightarrow ^6H_{15/2}$ transitions of Dy³⁺ ion reveal that the local environment around the RE³⁺ ion changed after the γ -ray irradiation in the present system. The lifetime of excited states of RE³⁺ ions decreased after the γ -ray irradiation due to the formation of defects induced by the γ -ray irradiation. The CIE color coordinates were determined before and after the γ -ray irradiation for the Dy³⁺-doped glass system.

Keywords Fluorophosphate glasses · Gamma-ray irradiation · Optical absorption and EPR spectroscopy

1 Introduction

Interest in radiation-induced defect formation in glasses has been increased due to the intensified application of high power lamps and lasers in optic systems [1–7]. In general, the interaction of high energetic radiation (laser, X-rays, γ -rays and others) with glasses is expected to produce

induced defects. Radiation energy higher than the band gap of the material potentially ionizes the electrons from their valence band. These free electrons and the resulting positive holes are captured by the already present intrinsic defects in glasses like vacancies, interstitial atoms, multivalent impurities and non-bridging oxygens, leading to the formation of electron defect centers (EC) and hole defect centers (HC) [5–15]. Therefore, the study of radiation-induced defect center formation in the glasses is of practical importance due to the application of high power lamps and lasers in optic systems. In particular, phosphate and fluorophosphate (FP) glasses are attractive candidates for high performance optics and optoelectronics, as well for lasers and amplifiers doped with rare earth ions because of their high transmission from the deep ultraviolet (UV) region to near infrared (NIR) region [5–7, 9, 10, 16–20]. On the other hand, these glasses are also suitable for the examination of the absorption and emission properties of impurities such as transition metal ions and rare earth ions as well as radiation induced defects [7]. Furthermore, FP glasses are also useful for developing a potential high-dose and high-resolution detector for microbeam radiation therapy [1, 2]. Under irradiation of FP and phosphate glasses with high energetic radiation,

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they solarize, that means, their transmission in the UV–vis regions decreases [18–20]. The radiation induced defect formation in these glasses was reported in detail by many authors [5–7, 9, 10, 16–20]. Previous studies have reported that both the fluoride and phosphate related defects can be formed by irradiation in FP glasses. It was also observed that the phosphate component of the FP glasses influenced more on the defect formation, due to higher stability of the phosphate-related defects compared with fluorine-related defects [7]. Griscom et al. reported that these defects in phosphate glasses are paramagnetic and their nature and structure were identified using electron paramagnetic-resonance (EPR) techniques [6]. However, many variations in the radiation induced defects are expected depending on the type and composition of glass, dose of irradiation and possible formation of electron/hole pairs.

It is well known that, in many host glasses, trivalent rare-earth ions such as Eu^{3+} and Sm^{3+} undergo redoxreaction that means they can be converted to their divalent form (Eu^{2+} and Sm^{2+}) upon exposure to high energetic radiation, such as X- and γ -rays because of their electronic structure and intense charge-transfer (CT) and 4f–5d transitions in the UV range [1, 13, 21, 22]. This valence change can be optically detected because the dominant emission bands of trivalent and divalent forms of Sm and Eu can be readily distinguished [1, 3, 13, 23]. All these properties make Eu and Sm ions are suitable indicators for examination of radiation-induced defects in glasses.

Hence, in the present report, we investigate RE^{3+} (RE = Sm, Eu and Dy)-doped FP glasses ($41\text{P}_2\text{O}_5 \cdot 17\text{K}_2\text{O} \cdot 9\text{CaO} \cdot 8\text{Al}_2\text{O}_3 \cdot 24\text{CaF}_2 \cdot 1.0\text{RE}_2\text{O}_3$, mol%) and studied the effect of γ -ray irradiation on optical properties through absorption, emission and electron paramagnetic resonance techniques.

2 Experimental Details

FP glasses with the composition (mol%) of $41\text{P}_2\text{O}_5 \cdot 17\text{K}_2\text{O} \cdot 9\text{CaO} \cdot 8\text{Al}_2\text{O}_3 \cdot 24\text{CaF}_2 \cdot 1.0\text{RE}_2\text{O}_3$, where RE = Sm, Eu and Dy, labeled as PKCFASm10, PKCFAEu10, PKCFADy10, respectively, were prepared by a conventional melt quenching technique. High-purity reagent grade chemicals of $\text{Al}(\text{PO}_3)_3$, KPO_3 , CaCO_3 , CaF_2 , and RE_2O_3 were used as starting materials to prepare the FP glasses. About 20 g of batch materials were thoroughly mixed and crushed in an agate mortar and the homogeneous mixture was transferred into a platinum crucible and kept in an electric furnace at 1150 °C for 1 h and then poured onto a preheated brass mold at 400 °C. Then the fabricated glasses were annealed at temperature which is close to the glass transition temperature (450 °C) for 12 h in order to remove strain and stress inside the glass samples. Finally, the prepared glasses were

polished with alumina and cerium oxide powders to achieve optical quality for optical and radiation measurements.

RE^{3+} (RE = Sm, Eu and Dy)-doped FP glasses were irradiated with a total dose of 48.56 kGy using γ -rays from a Co^{60} radiation source (MSD Nordion, pencil type/C-188 sealed). For all non-irradiated and irradiated samples, the absorption spectra were measured using a Cary 5000 UV–vis–NIR spectrophotometer with spectral resolution of 1 nm; electron paramagnetic resonance (EPR) spectra were measured using a JEOL Model JES FA200 instrument in X-band (9.450 GHz) at room temperature with a 100 kHz frequency modulation and 1 mW microwave power. The emission, excitation, and decay measurements were carried out by using Edinburgh FLS980 fluorescence spectrometer with a xenon arc lamp as an excitation source before and after the γ -irradiation.

3 Results and Discussion

3.1 Absorption and EPR Spectra

Figure 1 shows the optical absorption spectra of the RE^{3+} (RE = Sm, Eu and Dy)-doped FP glasses in UV–vis–NIR region before and after the γ -ray irradiation. It was found that the as prepared RE^{3+} (RE = Sm, Eu and Dy)-doped FP glasses are colorless but after the γ -ray irradiation the glass samples appeared brown colour as shown in inset of Fig. 1. As is shown in Fig. 1a, it can be found that Sm^{3+} -doped FP glasses exhibited totally 13 bands originating from ground state ($^6\text{H}_{5/2}$) to an excited states peaked at 344, 361, 374, 402, 437, 473, 942, 1077, 1229, 1373, 1482, 1530, and 1947 nm which are assigned to the $^6\text{H}_{5/2} \rightarrow ^4\text{D}_{7/2}$, $^4\text{D}_{3/2}$, $^6\text{P}_{7/2}$, $^6\text{P}_{3/2}$, $^4\text{F}_{5/2}$, $^4\text{I}_{9/2}$, $^6\text{F}_{11/2}$, $^6\text{F}_{9/2}$, $^6\text{F}_{7/2}$, $^6\text{F}_{5/2}$, $^6\text{P}_{3/2}$, $^6\text{F}_{1/2}$ and $^6\text{H}_{13/2}$ transitions, respectively.

In the case of Eu^{3+} -doped FP glasses, absorption bands in the UV–vis region centered at 362, 376, 382, 393, 465, and 525 nm which are ascribed to the $^7\text{F}_0 \rightarrow ^5\text{D}_4$, $^5\text{G}_3$, $^5\text{G}_2$, $^5\text{L}_6$, $^5\text{D}_2$, and $^5\text{D}_1$ transitions, respectively. Some bands centered at 413 and 535 nm are assigned to the $^7\text{F}_1 \rightarrow ^5\text{D}_3$ and $^5\text{D}_1$ transitions, respectively. Absorption bands appeared in NIR region correspond to $^7\text{F}_0 \rightarrow ^7\text{F}_6$ (2087 nm) and $^7\text{F}_1 \rightarrow ^7\text{F}_6$ (2208 nm) transitions (see Fig. 1b). Figure 1c shows the absorption spectra of Dy^{3+} -doped FP glasses before and after the γ -ray irradiation. The absorption bands of Dy^{3+} ions are due to the transitions from the $^6\text{H}_{15/2}$ ground state to the various excited states of Dy^{3+} ion, and were centered at 1678 ($^6\text{H}_{11/2}$), 1270 ($^6\text{H}_{9/2}$), 1093 ($^6\text{F}_{9/2}$), 896 ($^6\text{F}_{7/2}$), 750 ($^6\text{F}_{3/2}$), 473 ($^4\text{F}_{9/2}$), 453 ($^4\text{I}_{15/2}$), 424 ($^4\text{G}_{11/2}$), 386 ($^4\text{F}_{7/2}$), 363 ($^4\text{P}_{3/2}$) and 349 nm ($^6\text{P}_{7/2} + ^4\text{M}_{15/2}$).

It was noticed that the observed peaks were found to be very similar to that before γ -ray irradiation for all the studied FP glasses doped with RE^{3+} ions indicating that γ -ray

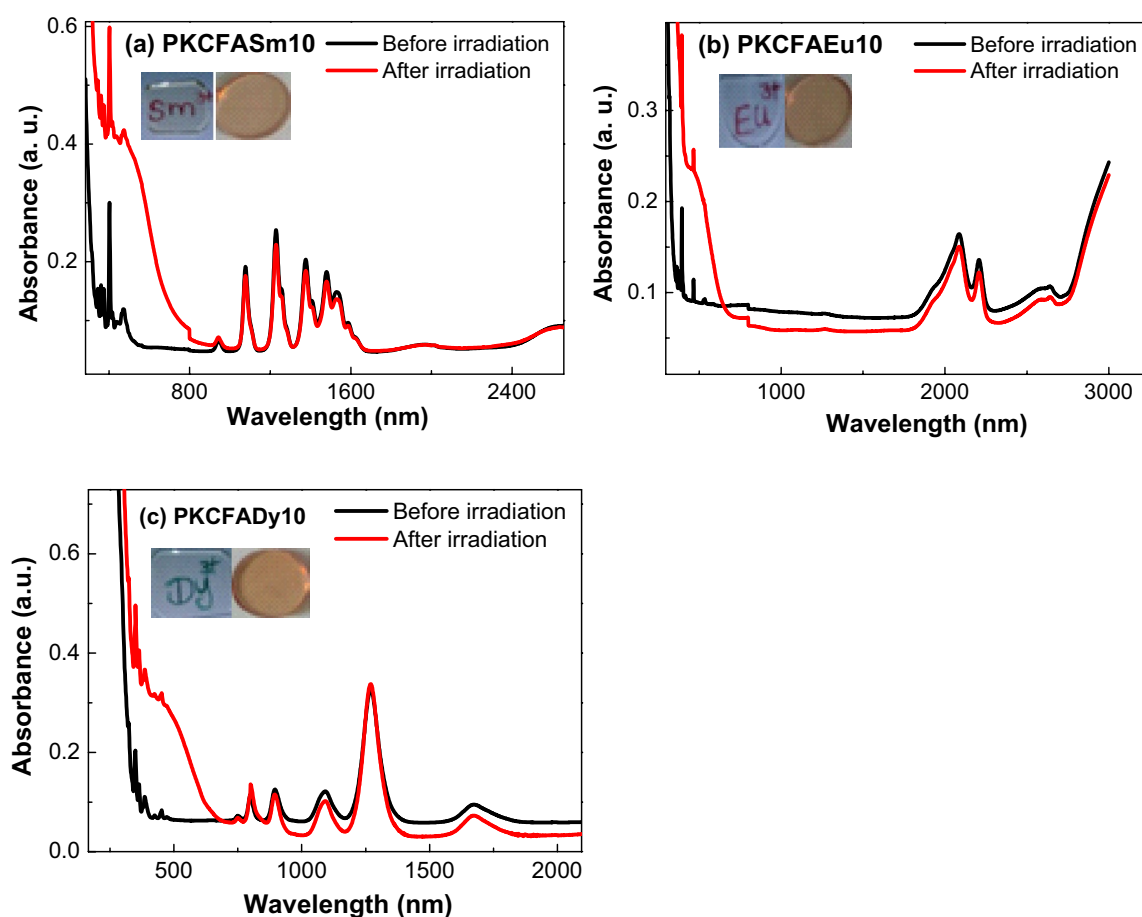


Fig. 1 Absorption spectra of the RE³⁺ (RE = Sm, Eu, Dy)-doped FP glasses before and after the γ -ray irradiation

irradiation causes no distinct variation in the entire absorption region. However, the absorption of the studied glasses increased at UV and visible regions after the γ -ray irradiation may be due to the defect centers induced during the γ -ray irradiation as reported elsewhere in phosphate glasses [7, 24]. In order to know the defect centers, the electron paramagnetic resonance (EPR) spectra were measured for all the γ -ray irradiated glass samples and are shown in Fig. 2. In the EPR spectra, for the irradiated glass samples, two main signals were observed due to ³¹P hyperfine splitting of 3.5 and 72 mT originate from phosphate-related defect centers, i.e. phosphorus-related non-bridging oxygen hole centers (POHC) and phosphorus-oxygen electron centers (POEC, trapped on the central phosphorus atom in PO₃ group). These two types of defect centers are characteristic of irradiated phosphate glasses as reported elsewhere [6, 7, 10, 25].

The nephelauxetic ratio β , and bonding parameter, δ , is defined as $\beta = v_c/v_a$ and $\delta = (1 - \bar{\beta})/\bar{\beta} \times 100$, where $\bar{\beta} = \sum_N \beta/N$, v_c and v_a are the energies of the corresponding transition in the complex and aquo-ions, respectively; 'N' refers to the number of aquo levels used to

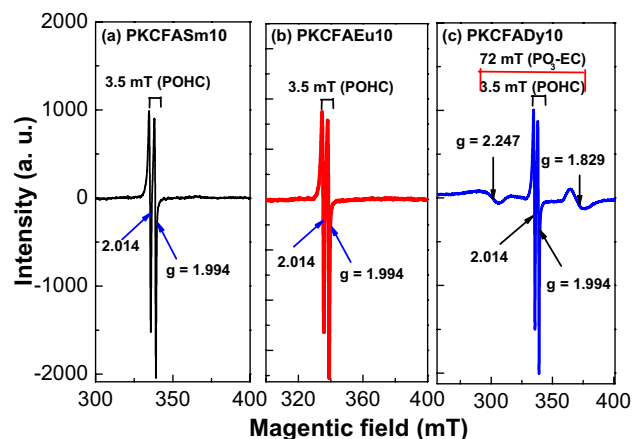


Fig. 2 Electron paramagnetic resonance (EPR) spectra of RE³⁺ (RE = Sm, Eu, Dy)-doped glasses irradiated with γ -rays

compute $\bar{\beta}$ values; 'δ' may be positive or negative indicating covalent or ionic bonding. In the present system, $\bar{\beta}$

and δ values were found to be 1.007 and -0.0069 for Dy^{3+} and 1.009 and -0.0093 for Sm^{3+} , which indicates that the RE-ligand bonding in the glass system is of ionic type.

Optical band gap energy (E_{opt}) is an important parameter to describe the absorption effects of optical materials. The study of the fundamental absorption edge in the UV-region is a useful method for the investigation of optical transitions and electronic band structure in crystalline and non-crystalline materials. According to Tauc [26], in many amorphous materials, variation of absorption coefficient with photon energy could be discussed in three different regions. The first region is known with almost constant absorption due to exciton–phonon coupling. The second part which is also known as Tauc region corresponds to high absorption from which optical band gap could be calculated. This region is associated with inter-band transitions. The third region is an exponential one called Urbach region. The details of the calculations of the optical band gap are described elsewhere [26, 27]. The E_{opt} for Eu, Sm and Dy ions before the γ -ray irradiation (after the γ -ray irradiation) was found to be 4.25 (4.0), 3.58 (4.25) and 3.58 (4.0) eV for indirect transitions, respectively. The changes in the optical band gap values are due to the formation of hole and electron defect centers and induced modifications in the energy levels due to γ -ray irradiation as evidenced from the EPR spectral analysis (see Fig. 2). The important feature of the absorption edge of amorphous materials is the exponential increase in the absorption coefficient with photon energy. With the energy of the incident photon less than the band gap, the increase in absorption coefficient follows with an exponential decay of density of states localized into the gap [28] and the absorption edge here is known as the Urbach energy. The Urbach energy indicates the width of the band tails of the localized states. The exponential absorption tails and the Urbach energy are connected by the relation [29]

$$\alpha(\nu) = \alpha_0 \exp\left(\frac{h\nu}{\Delta E}\right) \quad (1)$$

where α_0 is the constant and ΔE is the Urbach energy corresponds to the optical transition between the localized tail states adjacent to the valence band and the conduction bands which extend into the band gap. The Urbach energy value was found to be 0.29 (0.34), 0.28 (0.31) and 0.19 (0.29) eV for Eu, Sm and Dy ions before and after the γ -ray irradiation, respectively. The ΔE value indicates the disorderliness in the material and the changes in the values of ΔE are due to the formation of defects in the glasses. These defects produce localized states in the glasses causing the decrease in the width of the localized states in the optical band gap and in turn increase the optical band gap values.

3.2 Emission and Excitation Spectra

Figure 3a depicts the emission spectra of the Sm^{3+} -doped FP glasses before and after the γ -ray irradiation, obtained with an excitation of 404 nm. As can be seen, the spectra exhibited two strong and weak bands peaked at 561, 598, 645, and 705 nm which are assigned to the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{5/2}$, $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$, $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{9/2}$ and $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{11/2}$ transitions, respectively. Of the transitions, orange-reddish emission related to $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{7/2}$ transition (598 nm) was found to be stronger than the other transitions. Figure 3b illustrates the emission spectra of Eu^{3+} -doped FP glasses before and after the γ -ray irradiation, obtained by exciting the $^5\text{L}_6$ level of Eu^{3+} ion at 393 nm. It can be seen that the emission spectra consisted of five characteristic emission bands peaked at 578, 591, 611, 653, and 701 nm corresponding to $^5\text{D}_0 \rightarrow ^7\text{F}_0$, $^5\text{D}_0 \rightarrow ^7\text{F}_1$, $^5\text{D}_0 \rightarrow ^7\text{F}_2$, $^5\text{D}_0 \rightarrow ^7\text{F}_3$ and $^5\text{D}_0 \rightarrow ^7\text{F}_4$ transitions, respectively. Of the transitions, the $^5\text{D}_0 \rightarrow ^7\text{F}_2$ (611 nm) electric-dipole transition is hypersensitive which depends on ligand environment in the vicinity of Eu^{3+} ions while $^5\text{D}_0 \rightarrow ^7\text{F}_1$ magnetic-dipole transition hardly varies with the crystal-field strength around the Eu^{3+} ions. Therefore, intensity parameter R , which is the ratio of the intensities of the $^5\text{D}_0 \rightarrow ^7\text{F}_2/{}^5\text{D}_0 \rightarrow ^7\text{F}_1$ transitions, gives a measure of the degree of distortion from the inversion symmetry of the local environment of the Eu^{3+} ion in the lattice. Higher the value of R , lower the symmetry around the Eu^{3+} ions and higher the Eu–O covalence whereas lower the value of R , higher the symmetry around the Eu^{3+} ions and lower the Eu–O covalence [30]. From the emission spectra (Fig. 3b), the intensity parameter, R , was determined to be 3.97 and 4.41 before and after the γ -ray irradiation, respectively, which resembles that the Eu^{3+} -doped FP glasses exhibited lower symmetry around the Eu^{3+} ions and higher Eu–O covalency after the γ -ray irradiation.

Figure 3c shows the emission spectra of the Dy^{3+} -doped FP glasses before and after the γ -ray irradiation, obtained using the broadband source at 386 nm. The emission spectra consisted of blue and intense yellow bands, as well as a very weak red band, centered at 482, 574, 664, and 755 nm corresponding to $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$, $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$, $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{11/2}$, and $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{9/2}$, transitions of Dy^{3+} ion, respectively. Among the four bands, the blue band at 482 and yellow band at 574 nm were stronger than the other bands. The intensity of yellow band (electric-dipole transition) is known to be strongly influenced by the local environment of Dy^{3+} ions [31], in comparison to less sensitive magnetic-dipole transition (blue band) of Dy^{3+} . It is worth noting that there is a significant change in the relative intensity of the emission bands after the γ -ray irradiation. It results in different Y/B luminescence intensity ratios. The Y/B ratios were calculated for the 1.0 mol% Dy_2O_3 doped FP glasses before and after γ -ray irradiation and found to be 3.49 and

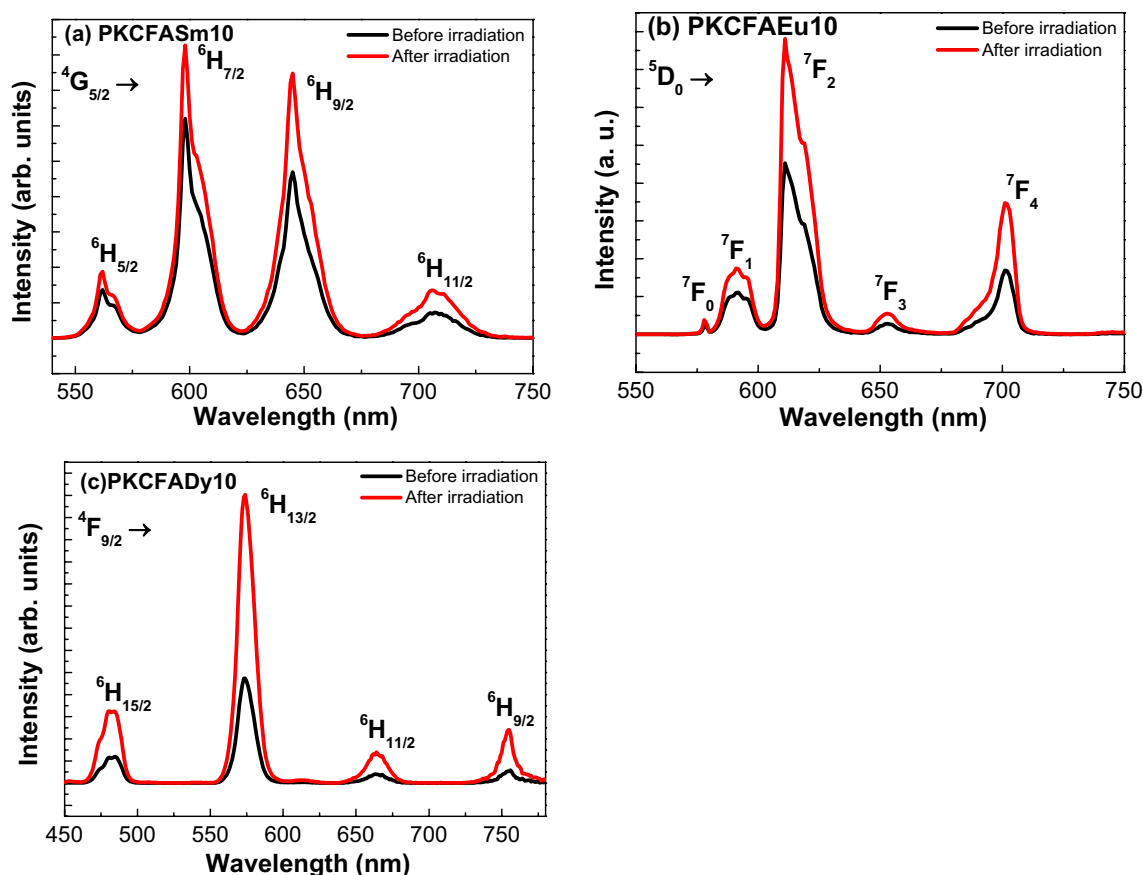


Fig. 3 Emission spectra of the RE³⁺ (RE = Sm, Eu, Dy)-doped FP glasses before and after the γ -ray irradiation

3.38, respectively. This change in the Y/B ratio is attributed to the change in the environment of Dy³⁺ ions in the glass sample as it involves a hypersensitive transition (${}^4F_{9/2} \rightarrow {}^6H_{13/2}$ with $J=2$).

Figure 4a presents the excitation spectra of Sm³⁺-doped FP glasses before and after the γ -ray irradiation, obtained by monitoring the emission at 598 nm. The spectra consisted of excitation bands in the 300–570 nm range at 344, 361, 374, 390, 402, 416, 440, 462, 473, 489, 500, 527, and 562 nm which are due to the transition from ${}^6H_{5/2}$ to various excited states of Sm³⁺ ion. As can be seen from the figure, the excitation bands in the range from 450 to 500 nm were due to closely spaced energy levels of Sm³⁺ ion located at these wavelengths. Figure 4b shows the excitation spectra of Eu³⁺-doped FP glasses before and after the γ -ray irradiation, obtained by registering the emission at 611 nm. The excitation spectra consist of the bands in the 300–550 nm spectral range at 303, 362, 366, 376, 382, 393, 415, 464, 525, and 533 nm which are due to the characteristic $4f^6-4f^6$ transitions of Eu³⁺ ions between the ground and excited levels. Of the bands, the bands at 393 and 464 nm corresponding to ${}^7F_0 \rightarrow {}^5L_6$ and ${}^7F_0 \rightarrow {}^5D_2$ transitions were relatively stronger intensity, indicating that violet and blue light emitting diodes

are suitable pumping sources to obtain Eu³⁺ emissions. Figure 4c depicts the excitation spectra of the Dy³⁺-doped FP glasses before and after γ -ray irradiation, obtained by registering the emission at 574 nm. There were several excitation bands in the region 300–500 nm, located at 324, 350, 364, 425, 453, and 473 nm, correspond to ${}^6H_{15/2} \rightarrow {}^6P_{3/2}$, ${}^6H_{15/2} \rightarrow {}^6P_{7/2}$, ${}^6H_{15/2} \rightarrow {}^4D_{5/2}$, ${}^6H_{15/2} \rightarrow {}^4G_{11/2}$, ${}^6H_{15/2} \rightarrow {}^4I_{15/2}$, and ${}^6H_{15/2} \rightarrow {}^4F_{9/2}$ transitions of the Dy³⁺ ion, respectively (Fig. 4c). Among these excitation bands, the one at 350 nm (${}^6H_{15/2} \rightarrow {}^6P_{7/2}$) was the most intense one which could be useful to get Dy³⁺ emissions. As shown in Fig. 4, the absorption of RE³⁺-doped FP glasses increased in the UV and visible regions after the γ -ray irradiation due to the defects formed by the γ -ray irradiation.

Phonon side band spectroscopy is one of the useful techniques to study the local structure surrounding the RE³⁺ ions in any host matrix. It is also used to determine the phonon energy, electron–phonon coupling strength of the host matrix and the phonon mode coupled to the RE³⁺ ions. These parameters of the host strongly affect the multiphonon relaxation processes. In the case of Eu³⁺ ion, the phonon side band (PSB) associated with the ${}^7F_0 \rightarrow {}^5D_2$ zero phonon (pure electronic) transition is clearly observed in the

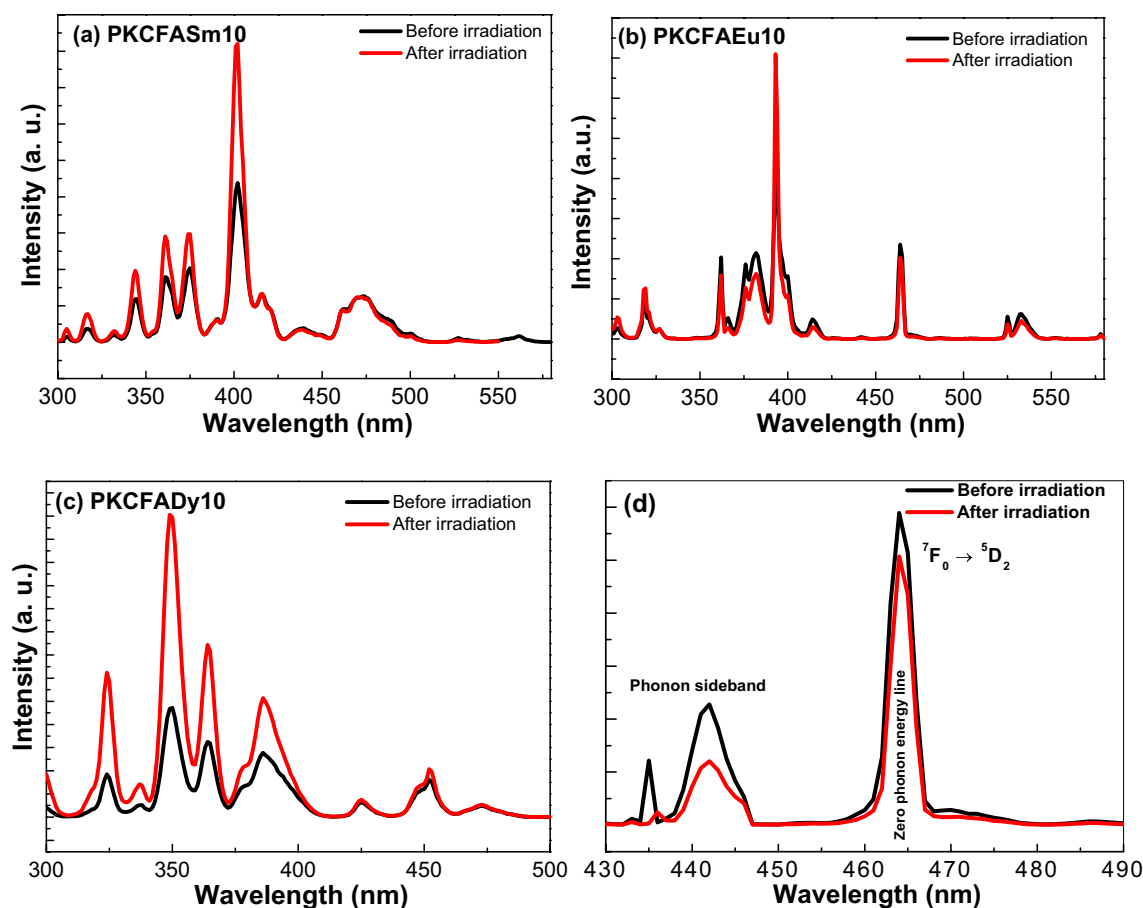


Fig. 4 Excitation spectra of (a) Sm^{3+} , (b) Eu^{3+} , (c) Dy^{3+} -doped FP glasses before and after the γ -ray irradiation and (d) phonon sideband spectra of Eu^{3+} -doped FP glass before and after the γ -ray irradiation

excitation spectrum, in the range of 430–480 nm, obtained by monitoring the emission at 611 nm. Figure 4d shows the phonon side band spectrum of Eu^{3+} -doped FP glass. The variation in the intensities of the phonon side band after the γ -ray irradiation is due to the change in the local environments of the Eu^{3+} ions. From the EPR spectral analysis, it was found that the phosphorus-related non-bridging hole centers (POHC) formed in Eu^{3+} -doped FP glasses upon exposure to γ -ray irradiation. These defect centers change the local environments of the RE ions and thus quenches the optical properties.

The intense band due to the pure electronic transition (PET), ${}^7\text{F}_0 \rightarrow {}^5\text{D}_2$, is located at 464 nm ($21,552\text{ cm}^{-1}$), while the PSB coupled to the ${}^7\text{F}_0 \rightarrow {}^5\text{D}_2$ transition of Eu^{3+} ions is observed in the higher-energy range at 441 nm ($22,676\text{ cm}^{-1}$). The difference between the position of the PSB and PET gives the phonon energy of the host which is found to be 1125 cm^{-1} in the present glass. The phonon sideband spectrum reveals that the intense phonon side bands are caused by the vibration with the greatest phonon energy. In FP glasses, phonons due to PO_2 groups in the metaphosphate

glasses mainly contribute to multiphonon relaxation and the electron–phonon coupling strength (g) is obtained from the phonon side band which is associated with the diphosphate groups. The ‘ g ’ is the relative measure of PSB line-strength and is given by $g = \int I_{\text{PSB}} d\nu / \int I_{\text{PET}} d\nu$ where I_{PSB} is phonon side band intensity and I_{PET} is the intensity of the pure electronic transition. The value of ‘ g ’ was found to be 0.5462 (0.378) before the γ -ray irradiation (after γ -ray irradiation). The lower ‘ g ’ value directly contributes to lesser non-radiative decay rates and thus enhances the radiative decay rates.

3.3 Decay Curves

Figure 5a shows the decay curves of Sm^{3+} -doped FP glasses before and after the γ -ray irradiation, obtained by monitoring the emission at 598 nm. As seen from figure, the decay curves exhibited non-exponential nature before and after the γ -ray irradiation. Figure 5b depicts the decay curves of Eu^{3+} -doped FP glasses before and after the γ -ray irradiation, obtained by registering the emission at 609 nm (${}^5\text{D}_0 \rightarrow {}^7\text{F}_2$ transition). As is seen,

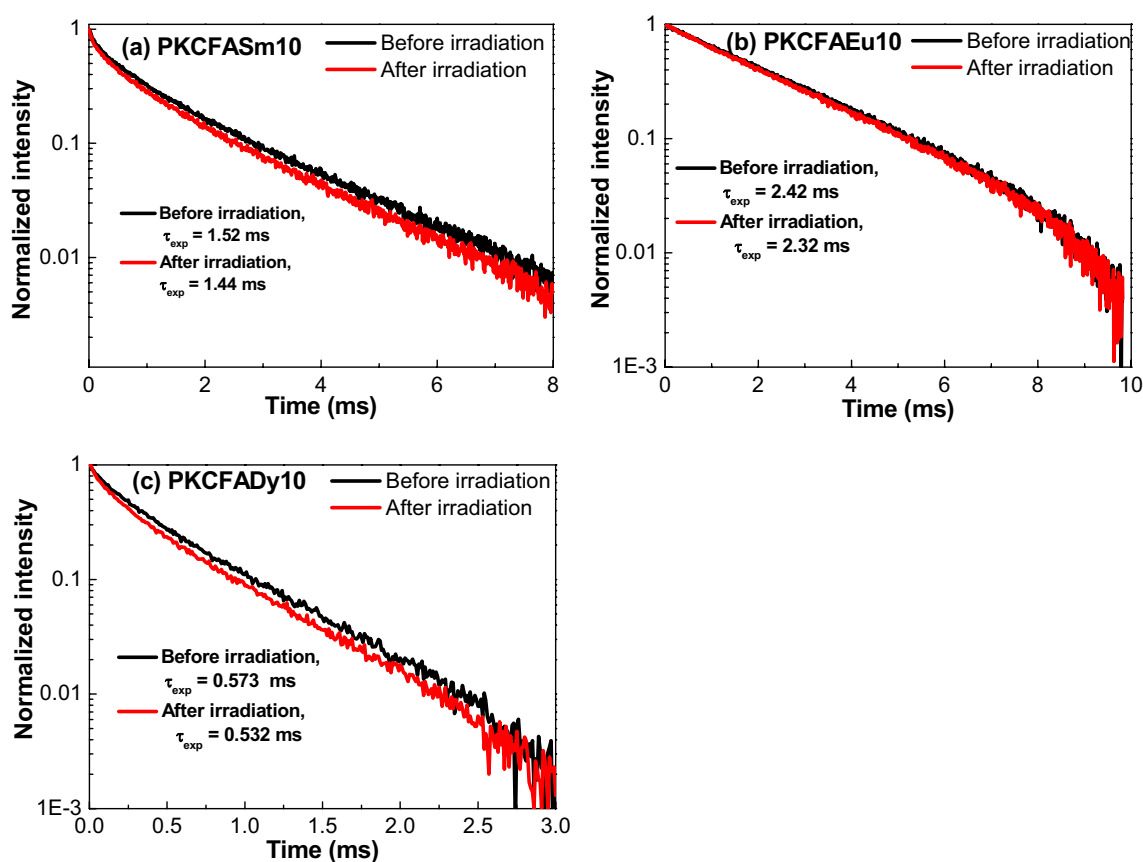


Fig. 5 Decay curves of RE^{3+} ($\text{RE} = \text{Sm}, \text{Eu}, \text{Dy}$)-doped FP glasses before and after the γ -ray irradiation

the decay curves exhibited single exponential nature before and after the γ -ray irradiation and lifetime was determined using the single exponential fitting. Figure 5c shows the decay curves for the $^4\text{F}_{9/2}$ level of Dy^{3+} ion before and after the γ -ray irradiation, obtained by registering the emission at 574 nm. It was found that the decay curves exhibited non-exponential nature regardless of the γ -ray irradiation. The effective lifetime (τ_{eff}) is considered as the experimental fluorescence lifetime (τ_{exp}), which is determined by using the expression [32]:

$$\tau_{\text{eff}} = \tau_{\text{exp}} = \frac{\int tI(t)dt}{\int I(t)dt} \quad (2)$$

The obtained τ_{exp} values of RE^{3+} -doped FP glasses before and after the γ -ray irradiation are shown in Fig. 5. In all samples, the lifetime was found to decrease after the γ -ray irradiation. This behavior can be understood by considering that the excited levels $^4\text{G}_{5/2}$ (Sm^{3+}), $^5\text{D}_0$ (Eu^{3+}), and $^4\text{F}_{9/2}$ (Dy^{3+}) are quenched by the defects induced by the irradiation.

3.4 Commission International de l'Eclairage Color Coordinates

The appropriate combination of blue and yellow emissions can produce white light [31, 33] and can find application as a potential solid state color display material in which generation and control of colors is possible [34, 35]. Dy^{3+} -doped glasses can be considered as attractive materials for two-colour phosphors and white light emission since Dy^{3+} ion exhibits intense emissions at blue (482 nm, $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$) and yellow (574 nm, $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$) regions. It is well known that the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{13/2}$ transition of Dy^{3+} ions is hypersensitive and its intensity strongly depends on the nature of the host, whereas the intensity of magnetic-dipole allowing the $^4\text{F}_{9/2} \rightarrow ^6\text{H}_{15/2}$ transition is less sensitive to the host. Therefore, the intensity ratio (Y/B) of these transitions can be tailored such that the Dy^{3+} -doped materials would generate white light. The white light produced by the emission of Dy^{3+} -doped glass samples can be analyzed in the frame of the chromaticity coordinates of colors. The procedure established by the Commission International de l'Eclairage (CIE)

is the most widely used method to describe the composition of any color in terms of three primaries colors i.e. Red, Green and Blue (RGB). Artificial “colors”, denoted by X, Y, Z, also called tristimulus values, can be added to produce real spectral colors (RGB). The chromaticity color coordinates is obtained from the following equations:

$$x = \frac{X}{X + Y + Z} \quad (3)$$

$$y = \frac{Y}{X + Y + Z} \quad (4)$$

and

$$z = \frac{Z}{X + Y + Z} \quad (5)$$

where X, Y, and Z are the tristimulus values given by: $X = \int E(\lambda)\bar{x}(\lambda)d\lambda$, $Y = \int E(\lambda)\bar{y}(\lambda)d\lambda$ and $Z = \int E(\lambda)\bar{z}(\lambda)d\lambda$. $\bar{x}$, $\bar{y}$, and $\bar{z}$ are matching functions of CIE, to define the chromatic characteristics of a visible source and they can be thought of as the spectral sensitivity curves of three linear light detectors yielding the CIE tristimulus values X, Y and Z.

The color coordinates for the Dy³⁺-doped FP glasses were calculated before and after the γ -ray irradiation and found to be $x=0.42$, $y=0.46$ and $x=0.42$, $y=0.46$, respectively and shown in the CIE (Commission Internationale de l'Éclairage) 1931 x – y chromaticity diagram (see Fig. 6). As can be seen, the x - and y -coordinates of color for the samples appeared near to the white light zone. It is also worth mentioning that no considerable variation has been found in the color coordinates after the γ -ray irradiation. The color coordinates of the present PKCFADy10 glass were found to be $x=0.42$, $y=0.46$ and were compared with

other Dy³⁺-doped glasses which include BN15 (0.40, 0.39) [13], PbPKANDy10 (0.45, 0.46) [36], LABD4 (0.34, 0.38) [37], LGBiBDy (0.37, 0.41) [38]. Thus, the present glasses may be considered as promising material for the white light applications.

4 Conclusions

Effect of the γ -ray irradiation on optical properties of the RE³⁺ (RE = Sm, Eu and Dy)-doped fluorophosphate glasses was investigated. From the EPR spectral analysis, it was found that the irradiated RE³⁺-doped fluorophosphate glasses exhibited different radiation-induced defect centers of POHC (phosphorus-related non-bridging oxygen hole centers), and PO₃ EC (electron trapped centers on the central phosphorus atom in the PO₃ group). From the absorption spectra, the bonding parameter and optical band gap values were evaluated before and after the γ -ray irradiation for all the studied systems. The bonding parameter confirms the ionic nature between the rare earths and ligand. The optical band gap energy (E_{opt}) and Urbach energy (ΔE) were found to increase after the γ -ray irradiation. The emission measurements demonstrate that the intensity of the emission bands significantly increased after the γ -ray irradiation. The intensity parameter, R, in Eu³⁺-doped fluorophosphate glasses confirms the higher covalency between Er–O bond and asymmetry around the Eu³⁺ ion after the γ -ray irradiation. The Y/B intensity ratio reveals that the local environment around the Dy³⁺ ions changed after the γ -ray irradiation due to the defect centers. The decay curves of the RE³⁺-doped fluorophosphate glasses were analyzed and found that the lifetime of the excited states of respective RE³⁺ ions decreased after the γ -ray irradiation. This behavior can be understood by considering that the excited levels ⁴G_{5/2} (Sm³⁺), ⁵D₀ (Eu³⁺), and ⁴F_{9/2} (Dy³⁺) are quenched by the defects induced by the irradiation.

Acknowledgements This work was partially supported by the New Growth Engine Industry Project of the Ministry of Trade, Industry and Energy, Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2013R1A1A2063250), and KEPKO Research Institute (KEPRI) and managed by KESRI (Project Number: KEPRI-16-23), South Korea. Dr. Ch. Basavapoornima is thankful to University Grants Commission, New Delhi, for the award of Post-Doctoral Fellowship for Women for the year of 2011–12 (F.15-1/2011–12/PDFWM-2011–12-OB-AND-9964 (SA-II), dt. 1-11-2013).

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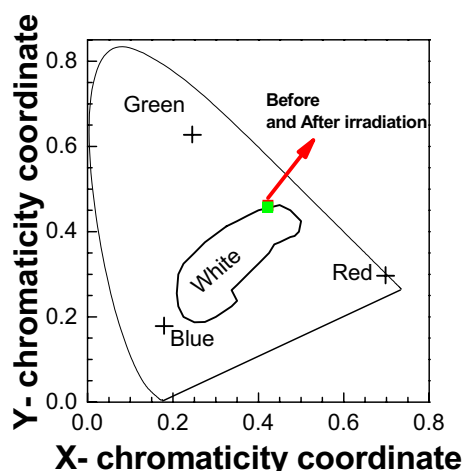


Fig. 6 CIE color diagram of Dy³⁺-doped FP glass before and after the γ -ray irradiation

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